Tara P. Mishra

2026 October 17

Curriculum Vitae

Max Planck Institute for Solid State Research
Stuttgart Center for Electron Microscopy (StEM),
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Research Areas

1. Atomistic engineering of next-generation functional materials through correlative quantitative electron microscopy integrated with multiscale computational modeling.
2. Advancing high-throughput, quantitative, and self-driving electron microscopy for accelerated materials characterization.

Professional Preparation

May 2026-Present	Max Planck Institute for Solid State Research	Group Leader (2-year contract)
Jan 2025-April 2026	Lawrence Berkeley National Lab (U.S.A.)	Project Scientist
July 2022-Dec 2024	Lawrence Berkeley National Lab (U.S.A.) Advisor: Prof. Gerbrand Ceder.	Postdoctoral Scholar
Aug 2018 - Apr 2022	National University of Singapore (Singapore) Singapore-MIT Alliance for Research and Technology Advisors: Profs. Canepa, Pennycook and Gradečak	Ph.D. Materials Science
2014-2018	Indian Institute of Science (India)	B.Sc., Research
May 2017-August 2017	National University of Singapore (NUS) Advisor: Prof. Stephen Pennycook	Summer Internship
May 2016-August 2016	Japan Advanced Institute of Science and Technology (JAIST) Advisor: Prof. Yoshifumi Oshima	Summer Internship

Research Experience

1. **Group Leader, MPI-FKF**
   - ▷ Extended NIMBUS to vendor-agnostic high-throughput and self-driving operation of electron-microscopes.
   - ▷ Federated deployment of the NIMBUS platform at MPI-FKF and Stanford University. Capability extension of the platform at Lawrence Berkeley National Laboratory.
2. **Project Scientist Experience**
   - ▷ Development of self-driving autonomous microscopy platform for new and legacy transmission electron microscopes.
   - ▷ Development of a clean transfer platform between AP-XPS and GIXS beamline and TEM.

- ▷ Understanding the transformation of earth-abundant layered cathodes by electrochemical pulsing using 4D-STEM techniques such as precession electron diffraction and electron ptychography.
- ▷ Development of standard operating procedures and training new users for automated microscopy procedures at the NCEM user facility.
- ▷ Co-supervision of 2 doctoral students and supervision of 1 undergraduate research student.

3. **Postdoctoral Experience**
   - ▷ Characterized and developed earth abundant cathode materials for all solid state batteries through electron microscopy techniques (atomic-resolution HAADF, 4D-STEM, EELS, EDS), including cryo conditions.
   - ▷ Automated electron microscopy and material characterization sample prep for self-driving labs.
   - ▷ Developed feature in py4DSTEM package for automated multiphase detection technique from 4D-STEM dataset.
4. **Doctor of Philosophy**
   - ▷ Decoded the origin of the compositional fluctuations in InGaN light emitting diodes (LEDs) using atomically resolved electron energy loss spectroscopy and multiscale computational modeling.
   - ▷ Developed an alternative pathway for manufacturing higher In-content defect-free InGaN LEDs through strain engineering.
   - ▷ Conducted fundamental investigations of sodium-ion transport in NaSICON superionic conductors using first-principles computational modeling, and developed an all-Python kinetic Monte Carlo software package to model ionic transport.

Honors & Awards

Oct 2025	Poster Award - 2nd Prize at 248th ECS meeting, Chicago
2023 to 2026	Elected Member of the Users Executive Committee for the Molecular Foundry.
June 2024	Berkeley Lab spot award for: "The identification of the δ-phase in DRX materials through advanced microscopy"
May 2018	Institute Gold Medal for best performance in Materials, IISc Bangalore
April 2018	Singapore-MIT Alliance for Research and Technology (SMART) fellowship. Fellowship to pursue Ph.D. under supervision of Professors from NUS and MIT.
August 2017	Most Promising Young Researcher in Materials Students' Symposium, IISc Bangalore.
May 2016	JASSO (Japan Student Services Organization) scholarship for summer internship at Japan Advanced Institute of Science and Technology (JAIST)
March 2014	University Entrance Exam (KVPY) All India Rank: 105

Skills

- ▷ Quantitative materials characterization using electron microscopy techniques (atomic-resolution HAADF-STEM, 4D-STEM, and spectroscopy) under different environmental conditions such as cryo and in-situ.
- ▷ Transmission electron microscopy sample preparation of battery and semiconductor materials using focused ion beam (FIB), polishing, and drop casting.
- ▷ Materials modeling using first-principles, molecular dynamics, and Monte Carlo methods.
- ▷ Image and signal processing toolkit development for quantitative information extraction from electron microscopy measurements.
- ▷ Proficient in programming languages such as Python, C++, C, and MATLAB.
- ▷ Experienced in materials simulation packages such as VASP, LAMMPS, pymatgen, CRYSTAL, and CASM.
- ▷ Experienced in electron microscopy simulation and data analysis packages such as abTEM, py4DSTEM, and HyperSpy.
- ▷ Maintenance and usage of Linux-based high-performance computing cluster.
- ▷ Development of automated material characterization platforms involving electron microscopes.

Open source package development

- ▷ **Lead developer of NIMBUS:** A vendor-agnostic self-driving TEM platform; deployed at MPI-FKF, LBNL, and Stanford. Github source.
- ▷ **Contributor to py4DSTEM:** An open source set of python tools for processing and analysis of four-dimensional scanning transmission electron microscopy (4D-STEM) data. Github source.
- ▷ **Contributor to HyperSpy:** open source Python library for the interactive analysis of multidimensional datasets that can be described as multidimensional arrays of a given signal. Github source.

Publications

*: co-first author and #: Corresponding author

Under-review publications

24. **T. P. Mishra***, Z. Li*, M. Shen, M. Jaugstetter, L.P. Matte, J.O. Park, B. Lam, K. Bustillo, G. Ceder, M.C. Scott, *Investigating the Degradation of LATP Solid Electrolyte in High Alkaline* $Li - O_2$ *Batteries.* (Under review) arXiv:2508.08654

Peer-reviewed Publications

First, co-first, or corresponding author

23. E. Lee*, T. Holstun*, **T. P. Mishra***, M. Mozdzierz, S. Wang, P. Ercius, A. Pattison, S. Hiroi, K. Ohara, V. Battaglia, H. Kim, A. Miura, G. Ceder, *Near-micron single-crystalline Mn-rich cathodes with exceptional cycling stability.* (In press: Advanced Energy Materials) 10.26434/chemrxiv.15005574

22. N. Scherz, S. Anand, C. Ophus, T. Holstun, M. Scott, **T. P. Mishra**#, G. Ceder#, *Crystallographic challenges in microscopy of multi-domain spinel materials*. **Journal of Materials Chemistry A** (2026). 10.1039/D6TA05906F.

21. J.Y. Chung*, **T. P. Mishra***, Z. Mahfoud*, T. E. Gage, J. Wen, K. P. Rice, I. Martin, L. Zhang, G. J. Syaranamual, S. J. Pennycook, S. Gradečak, P. Canepa, M. Bosman, *Nanoscale Optical Inhomogeneities From Compositional Segregation Within Individual GaN-on-Si Quantum Wells*, **Advanced Science** (2026). 10.1002/advs.76414.

20. D. Milsted*, **T. P. Mishra***, G. Ceder, *Automated sample preparation system for electron microscopy characterization techniques*, **Digital Discovery** (2025). 10.1039/D5DD00116A.

19. T. Holstun*, **T. P. Mishra***, L. Huang, H.M. Hau, X. Yang, C. Ophus, K. Bustillo, L. Ma, S. Ehrlich, G. Ceder, *Accelerating the Electrochemical Formation of the δ Phase in Manganese-Rich Rocksalt Cathodes*, **Advanced Materials** (2024). 10.1002/adma.202412871.

18. H.M. Hau*, **T. P. Mishra***#, C. Ophus, T. Y Huang, K. Bustillo, Y. Sun, X. Yang, T. Holstun, X. Zhao, S. Wang, Y. Ha, G-H Lee, C. Song, J. Turner, J. Bai, L. Ma, K. Chen, F. Wang, W. Yang, B. D. McCloskey, Z. Cai#, G. Ceder#, *Earth-abundant Li-ion cathode materials with nanoengineered microstructures*, **Nature Nanotechnology** (2024). 10.1038/s41565-024-01787-y.

17. **T. P. Mishra**, G. J. Syaranamual, Z. Deng, J.-Y. Chung, L. Zhang, S. A. Goodman, L. Jones, M. Bosman, S. Gradečak, S. J. Pennycook, and P. Canepa, *Unlocking the origin of compositional fluctuations in InGaN light emitting diodes*, **Physical Review Materials** (2021). 10.1103/PhysRevMaterials.5.024605.

16. **T. P. Mishra**, M. Koyano, and Y. Oshima, *Detection of large thermal vibration for Cu atoms in tetrahedrite by high-angle annular dark-field imaging*, **Applied Physics Express** (2017). 10.7567/APEX.10.045601.

Significant contributions

15. S. Anand, **T. P. Mishra**, P. Zhong, K. Jun, T. Holstun, and G. Ceder, *Origin of enhanced performance when Mn-rich rocksalt cathodes transform to δ-DRX.* **Materials Today** (2025). 10.1016/j.mattod.2025.06.024

14. L.P. Matte, M. Jaugstetter, A. S. Thill, **T. P. Mishra**, C. Escudero, G. Conti, S. Nemsak, F. Bernardi, *Direct observation of phase change accommodating hydrogen uptake in bimetallic nanoparticles.* **ACS Nano** (2025). 10.1021/acsnano.4c18013

13. Z.Y. Deng, **T. P. Mishra**, E. Mahayoni, Q. Ma, A. J. K. Tieu, O. Guillon, J. N. Chotard, V. Seznec, A. K. Cheetham, C. Masquelier, G. S. Gautam, and P. Canepa, *Fundamental investigations on the sodium-ion transport properties of mixed polyanion solid-state battery electrolytes*, **Nature Communications** (2022). 10.1038/s41467-022-32190-7.

12. Y. Gao, **T. P. Mishra**, S. H. Bao, G. S. Gautam, and P. Canepa, *Design and Characterization of Host-Frameworks for Facile Magnesium Transport*, **Annual Review of Materials Research** (2022). 10.1146/annurev-matsci-081420-041617.

Contributing author

11. P. Zhong, B. Deng, S. Anand, **T. P. Mishra**, G. Ceder, *Modeling phase transformations in Mn-rich cathodes with charge-informed machine-learning interatomic potentials.* **Physical Review**

Materials, 2025 10.1103/mk2d-tjyj

10. J. Lee, L. P. Matte, Z. C. Tronstad, T. Holstun, **T. P. Mishra**, J. O. Park, J. W. Kim, J. R. Long, G. Ceder, M. C. Scott, B. D. McCloskey, *Understanding Electrochemistry of Humidified Solid-State Lithium-Oxygen Batteries.* **Advanced Energy Materials**, 2025 10.1002/aenm.202504050

9. G. Z. Girotto, M. Jaugstetter, D. Kim, L. P. Matte, **T. P. Mishra**, M. Scott, R. M. Martins, A. R. Muniz, M. Salmeron, S. Nemsak, F. Bernardi, *Shapeshifting Nanocatalyst for CO_2 Conversion.* **Advanced Materials** (2025) 10.1002/adma.202509814

8. L.N. Walters, B. Rendy, Y. Fei, M. Diallo, K. Jun, X. Yang, G. Wei, M. McDermott, **T. P. Mishra**, F. Shen, D. Milsted, M.S. Oo, H. Kim, M.C. Tucker, and G. Ceder, *Survey of the $Na_{8-x}A_xP_2O_9$ (NAP) sodium superionic conductor framework.* **Chemistry of Materials** (2025) 10.1021/acs.chemmater.5c01573

7. H. M. Hau, T. Holstun, E. Lee, B.L.D. Rinkel, **T. P. Mishra**, M.M. DiPrince, R.S. Mohankrishnan, E.C. Self, K.A. Persson, B. D. McCloskey, G. Ceder, *Disordered Rocksalts as High-Energy and Earth-Abundant Li-Ion Cathodes* **Advanced Materials** (2025). 10.1002/adma.202502766

6. B. Lam, Z. Li, **T. P. Mishra**, and G. Ceder, *Degradation Mechanism of Phosphate-Based Li-NaSICON Conductors in Alkaline Environment.* **Advanced Energy Materials** (2024). 10.1002/aenm.202403596.

5. J.Y. Chung, Y. Yuan, **T. P. Mishra**, C. Joseph, P. Canepa, E. H. Sadki, S. Gradečak, and S. Garaj, *Structure and exfoliation mechanism of two-dimensional boron nanosheets*, **Nature Communications** (2024). 10.1038/s41467-024-49974-8.

4. Z. Wang, **T. P. Mishra**, W. Xie, G. S. Gautam, and P. Canepa, *Kinetic Monte Carlo Simulations of Sodium Ion Transport in NaSICON Electrodes*, **ACS Materials Letters** (2023). 10.1021/acsmaterialslett.3c00610.

3. Z. Deng, **T. P. Mishra**, W. Xie, D. A. Saeed, G. S. Gautam, and P. Canepa, *kMCpy: A python package to simulate transport properties in solids with kinetic Monte Carlo*, **Computational Materials Science** (2023). 10.1016/j.commatsci.2023.112394.

2. J-Y Chung, Z. Li, S. A. Goodman, J. So, G. J. Syaranamual, **T. P. Mishra**, E. A. Fitzgerald, M. Bosman, K. Lee, S. J. Pennycook, and S. Gradečak, *Light-Emitting V-Pits: An Alternative Approach toward Luminescent Indium-Rich InGaN Quantum Dots*, **ACS Photonics** (2021). 10.1021/acsphotonics.1c01009.

1. M. Li, Z. Huang, C. Tang, D. Song, **T. P. Mishra**, A. Ariando, T. Venkatesan, C. Li, S. J. Pennycook, *Correlated Lattice Instability and Emergent Charged Domain Walls at Oxide Heterointerfaces*, **Advanced Functional Materials** 29(49), 1906655 (2019) 10.1002/adfm.201906655.

Patents and invention disclosures

3. Synthesis of partially disordered phase in rocksalt materials via electrochemical pulsing. International application, WIPO; Publication No. WO/2025/075989.

2. High-capacity Mn-based rocksalt cathodes with structural changes. International application, WIPO; Publication No. WO/2025/188327.

1. Autonomous sample preparation for electron microscopy. (Invention disclosure filed with Lawrence Berkeley National Lab).

Grants and funding

2. Co-PI in Laboratory Directed Research & Development (LDRD) Program by Lawrence Berkeley National Lab for the development of "Bragg Enhanced Microsecond XAS for Real-time Reactions of Materials in Gas and Liquid Environments", Oct 2025. Total Funding: US$250,000.

1. Co-PI in Rapid Development of Instrumentation for us (Radius) by Advanced Light Source for the development of "Integrating X-ray characterization and electron microscopy by unified sample holder with clean transfer", Aug 2024. Total Funding: US$300,000.

Invited & conference presentations

12. *Understanding the atomic habits of materials for energy storage.* **Institute Colloquium, Max Planck Institute for Solid State Research, Stuttgart**, Oct 2026. (Invited Talk).

11. *Looking Closer, repeatedly, until Energy Storage Makes Sense.* **University of California, Santa Barbara**, Feb 2026. (Invited Talk).

10. **T. P. Mishra**, T. Holstun, S. Anand, K. Bustillo, C. Ophus, G. Ceder, *Development of next generation DRX cathodes through the lenses of electron microscopes.* **Electrochemical Society meeting, Chicago**, 2025. (Poster: 2nd Prize Winner)

9. **T. P. Mishra**, T. Holstun, H. M. Hau, C. Ophus, K. Bustillo, G. Ceder, *Multimodal electron microscopy characterization of electrochemically formed δ phase in Manganese-Rich Rocksalt Cathodes.* **Microscopy and Microanalysis, Salt Lake City**, 2025. (Oral Talk)

8. **T. P. Mishra**, H. M. Hau, T. Holstun, C. Ophus, K. Bustillo, G. Ceder, *Unraveling nano-engineered microstructure in large particle DRX cathode particles.* **MRS Fall Meeting, Boston, USA**, 2024. (Oral Talk)

7. **T. P. Mishra**, *Exploring next-generation DRX cathodes through multimodal electron microscopy.* **Molecular Foundry, Berkeley Lab, USA** 2024. (Invited talk).

6. **T. P. Mishra**, *Hands on tutorial on the automated orientation mapping module of the py4DSTEM package.* **HyperSpy workshop 2024, Diamond Light Source, UK**, 2024. (Invited Workshop)

5. **T. P. Mishra**, Z.Y. Deng, J.Y. Chung, S. Gradečak, S.J. Pennycook, and P. Canepa, *Detection and suppression of compositional fluctuations in InGaN Light Emitting Diodes.* **MRS Fall Meeting, Boston, USA**, 2021. (Oral Talk)

4. **T. P. Mishra**, Z.Y. Deng, J.Y. Chung, S. Gradečak, S.J. Pennycook, and P. Canepa, "Origin of Compositional fluctuations in InGaN Light Emitting Diodes." **Institute of Physics Singapore**, 2021. (Oral Talk)

3. **T. P. Mishra**, G. J. Syaranamual, L. Jones, J. Y. Chung, Z. Li, S. A. Goodman, S. J. Chua, E. A. Fitzgerald, P. Canepa, S. Gradečak, and S. J. Pennycook, "Quantitative Measurement of Sub-nanometer In Fluctuations in InGaN Quantum Well", **ICMASS, Nagoya**, 1392 (2019) (Poster)

2. **T. P. Mishra**, M. Koyano and Y. Oshima, "Detection of large thermal vibration for Cu atoms in tetrahedrite by high-angle annular dark-field imaging", **ICMAT**. (2019) (Poster)

1. **T. P. Mishra**, M. Koyano and Y. Oshima, "Anomalous scattering at high angles stemming from anisotropic atomic vibrations", **Materials Student Symposium, Indian Institute of Science, Bangalore** (2017). (Oral Talk)

Teaching and mentoring

1. **2026-Present** : Doctoral research co-supervision of Mr. Cedric Lim (Stanford/LBNL) for his research in "Development of advanced algorithms for material characterization."
2. **2023-2025** : Undergraduate research supervision of Ms. Ninon Odile Scherz (UC Berkeley/LBNL) for her research in "Crystallographic Challenges in Microscopy of Multidomain Spinel Materials."
3. **2022-Present** : Doctoral research co-supervision of Ms. Meghan Shen (UC Berkeley/LBNL) for her research in "Electron microscopy investigations in degradation of Mn-rich disordered cathode materials."
4. **Fall, 2021** Undergraduate research supervision of Mr. Preston Lim (NUS) for his Final Year Project on "Machine Learning of Interatomic Potentials in InGaN alloys".
5. **Fall, 2017** Teaching Assistant, Undergraduate Core Module, UMT 202: Structure of Materials at Indian Institute of Science.

Research in the news

1. **April 28, 2021** MIT News Office: SMART investigates the science behind varying performance of different colored LEDs. https://news.mit.edu/2021/smart-performance-different-colored-leds-0428
2. **Sept 25, 2024** News from Berkeley lab: Manganese Cathodes Could Boost Lithium-ion Batteries https://newscenter.lbl.gov/2024/09/25/manganese-cathodes-could-boost-lithium-ion-batteries/